Pragmatism, Not Ideology

Drivers of E-Bike Sharing Usage Intensity

Michael Stiebe (HSLU Luzern / University of Basel)
Benjamin Weggelaar (University of Basel)

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The Research Team

Michael Stiebe

HSLU

Competence Center for Mobility

Faculty of Business and Economics

Sustainable mobility systems
shared transport, e-bike adoption
urban mobility behavior

Benjamin Weggelaar

University of Basel

Faculty of Business and Economics

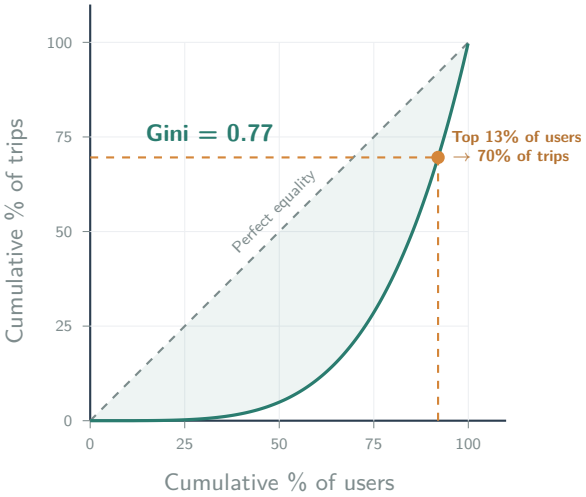
Environmental & transport economics
causal inference, survey methods
mobility behavior

With Frank Krysiak (University of Basel) & Widar von Arx (HSLU)

Contemporary EBSS Challenges

- **High operational costs & low WTP**
  cf. Reckermann et al., 2025; Bieliński et al., 2024
- **Geographic disparities:** usually urban-focused, suburban and peripheral regions rarely covered
- **Weather-sensitive/seasonal demand**
  cf. Noland, 2021; Reck, 2019
- **Adverse (PT) substitution effects**
  Bieliński et al., 2021; Reck & Axhausen, 2022
- **Demographic/socioeconomic inequality**
  (mainly male, highly educated, high income users)
  cf. Reck et al., 2021; Bielinski et al., 2021; Hess & Schubert, 2019

A handful of riders take most of the trips



E-bike sharing is driven by a small power user population.

- 1.4 million trips in Basel, 2018–2025
- Gini = 0.77 — much more than other transport modes
- **Top 13%** of users generate **70% of all trips**
- **Who are these power users?** And **why** do they ride so much?

What are key driver/inhibitors of EBSS usage?

The dominant narrative:

"Green attitudes drive green transport adoption."

What we find:

Intensive users are pragmatists, not environmentalists.

Our research question:

What distinguishes *intensive* e-bike sharing users from occasional users — attitudes, behaviors, or context?

Why it matters:

To better understand the role of EBSS in a sustainable transport network, it is necessary to identify the determinants of their usage.

This study links stated attitudes to *actual* administrative trip records

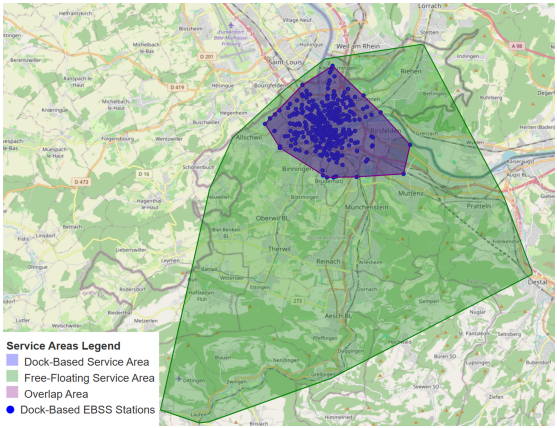
Dimension	Survey studies	Usage data studies
Behavior	Stated	Revealed
Time coverage	Cross-section	Time series
Sample scope	Small, self-selected	Full user population
Spatial resolution	Home/work zones	E-bike coordinates

Survey responses matched to actual riding behavior.

N = 1,743 survey respondents and 1.4M trip records ($\sim$ 10% linked to survey)

Combining survey responses to actual riding behavior is quite rare in transportation studies

Motivation



pick bike

PubliBike velospot

FF-EBSS in Basel since 2018

DB-EBSS in Basel since 2021

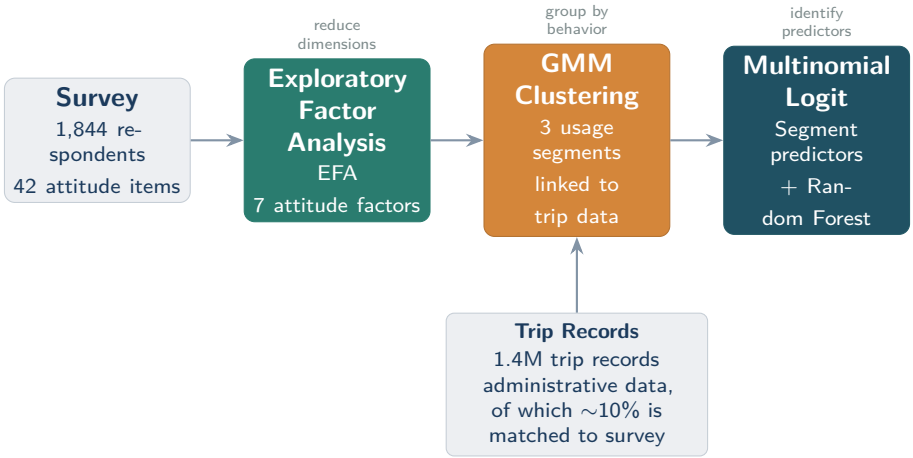
Current mobility system under pressure

Shared micromobility services gain traction

Basel as ideal testbed:

- Highest cycling modal share & lowest motorization rate among Swiss cities
- Dynamic tri-state area (CH/D/F)
- 2037 net-zero target
- Strong (but stressed) PT backbone
- EBSS as potential catalyst for sustainable inter/multimodal mobility

Three methods, one integrated picture



Survey: capturing the full attitudinal landscape

Survey instrument (April 2025)

Respondents:	1,844 (both systems)
Languages:	German, French, English
Items:	42 attitude items (5-pt Likert)
Domains:	Env. values, functional perceptions, multimodal use, weather attitudes, hedonic motivation, reliability concerns
Linkage:	Respondent ID matched to operator's trip database
Deployment:	Online, via operator invitation

Controls captured:

- Age, gender, employment status
- Car ownership, driver's license
- Public transport season ticket
- E-bike ownership
- Residential location (linked to geospatial data)

Sample Characteristics

Characteristic	Free-Floating (Pick-e-Bike) $n = 1{,}382$	Dock-Based (PubliBike) $n = 462$
Demographics		
Mean age (SD)	43.8 (14.0)	38.2 (12.2)
% Male	74.9%	64.7%
% Master's/Doctoral degree	34.1%	58.0%
Mobility Assets		
% Car ownership (personal)	61.2%	37.9%
% Private e-bike ownership	34.4%	16.2%
% Half-fare card	63.0%	58.2%
% GA travelcard	10.9%	21.6%
Geography		
% Lives in Basel city	31.0%	53.7%
% Urban residence	46.0%	74.6%
Usage Patterns		
Mean membership duration (months)	54.2	13.8
Mean rentals per month	1.75	4.30
% High-intensity users	9.6%	23.5%

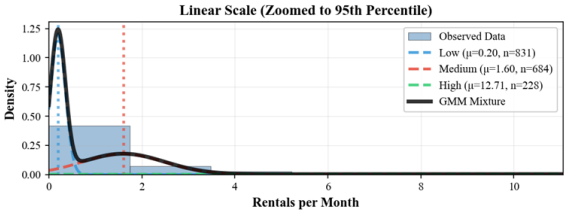
EFA: 42 Likert scale items reduce to 7 interpretable attitudinal dimensions

Factor	Label	α	Items	Description
F1	**Weather-Independent Usage**	0.813	6	Stated willingness to use EBSS under adverse weather conditions
F2	Public Sector Intervention Attitudes	0.532	5	Support for subsidies, infrastructure, policy integration
F3	Hedonic Motivation & Social Support	0.665	4	Perceived enjoyment, comfort, social endorsement
F4	**Perceived Functional Superiority**	0.728	5	Beliefs that EBSS is faster, more convenient than public transit
F5	**Multimodal Complementarity**	0.598	4	Combining EBSS with walking, transit, and mobility backup use
F6	Service Reliability & Affordability	0.401	4	Perceived cost and availability barriers *(low α; treat with caution)*
F7	**Environmental Self-Identity**	0.633	2	Environmental responsibility, EBSS sustainability beliefs

Extraction: principal axis factoring; Kaiser criterion (eigenvalue > 1); oblique rotation (promax). Standardized factor scores ($M = 0$, $SD = 1$) via regression method.

F1, **F4**, **F5**, **F7** are the key factors in the regression results.

GMM clustering: BIC selects 3 segments from behavioral trip data



Clustering approach:

- Applied to: trip frequency
- BIC selects $K = 3$; probabilistic membership

Segments:

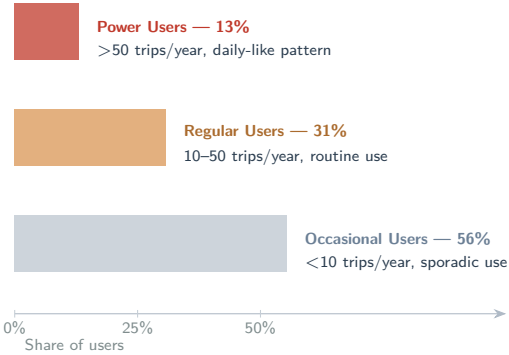
Power (>50 trips/yr) · **Regular** (10–50/yr) · **Occasional** (<10/yr)

System power user rates:

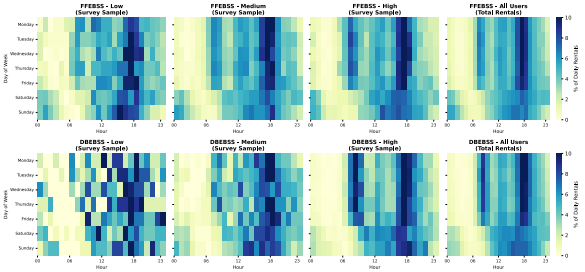
PubliBike (dock-based): **23.5%** power users

Pick-e-Bike (free-floating): **9.6%** power users

Usage falls into three distinct segments



When do users ride?



Power users show pronounced AM/PM commuter peaks.

Occasional users cluster at midday — leisure, not routine.

High-intensity users ride year-round; seasonal amplitude only 7.7pp vs. 12.3pp for low-intensity.

Multinomial logit: predicting segment membership from factor scores

Outcome: Segment $\in$ {Power, Regular, Occasional}
Reference: Occasional

Predictors (attitudinal, F1–F7):
F1 Weather F2 Intervention F3 Hedonic
F4 Functional superiority F5 Multimodal
F6 Reliability F7 Environmental identity
Controls: age, gender, employment, car, PT sub., system, distance

Validation: Random Forest
Variable importance confirms MNL coefficient ordering

Three model variants:
(1) Combined sample
(2) PubliBike only (3) Pick-e-Bike only

Results as Relative Risk Ratios (RRR $= \exp \beta$). RRR > 1: predicts more intensive use. RRR < 1: predicts less intensive use.

Results reported as RRR:

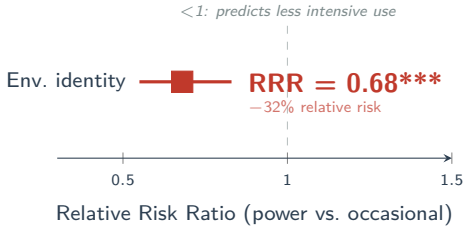
Relative Risk Ratios $= \exp(\beta)$
RRR > 1: factor predicts **more intensive** membership
RRR < 1: factor predicts **less intensive** membership

Why three models?

Free-floating and dock-based attract fundamentally different users. System-specific models reveal whether the drivers of intensity differ across EBSS types.

The combined model gives the aggregate picture. The system-specific models reveal where it breaks down.

Environmental identity predicts *less* intensive use



The greener the identity, the lighter the use.

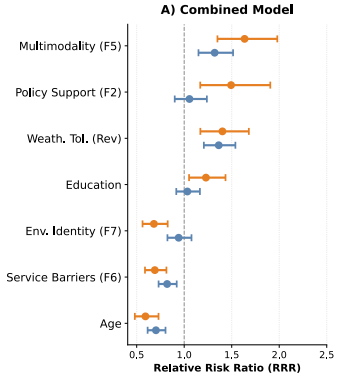
Higher environmental self-identity predicts being in the *occasional* rather than *power* user segment.

Why?

Environmentally committed users treat EBSS as one of *many* green choices — cycling, walking, transit. They do not rely on any single mode.

Power users are committed to **the bike**, not the environment.

Three pragmatic factors drive intensive use



Points = RRR; bars = 95% CI. Dashed line at RRR=1.

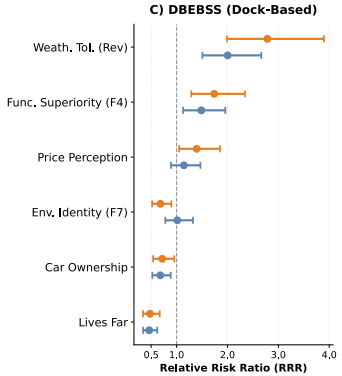
Medium Intensity High Intensity

Combined model — key predictors:

- **Weather tolerance**
  RRR = 1.40***
- **Multimodal integration**
  RRR = 1.64***
- **Functional superiority over PT**
  RRR = 1.74***
- **Environmental identity**
  RRR = 0.68*** (negative)

RF confirms variable importance ordering.

PubliBike (dock-based): weather resilience dominates



Points = RRR; bars = 95% CI. Dashed line at RRR=1.

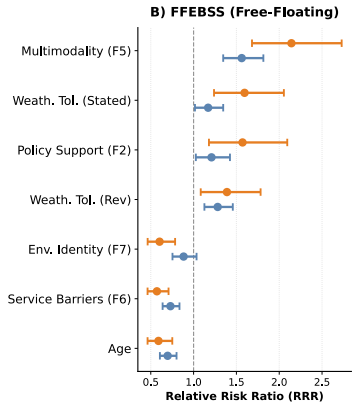
Medium Intensity High Intensity

PubliBike power user profile:

- **Weather resilience dominant** — RRR ≈ 2.78
- **Functional superiority** strong secondary driver
- Multimodal positive but secondary
- Env. identity: negative (RRR ≈ 0.63)

23.5% power users. Dock-based commuters lock in daily routes — weather tolerance is decisive.

Pick-e-Bike (free-floating): multimodal integration dominates



Points = RRR; bars = 95% CI. Dashed line at RRR=1.

Medium Intensity High Intensity

Pick-e-Bike power user profile:

- **Multimodal integration dominant** — RRR ≈ 1.90
- **Functional superiority** positive secondary
- **Weather is significant**
- Env. identity negative

Only **9.6%** power users. Free-floating fills last-mile gaps — not daily commuting.

System type fundamentally shapes the driver profile

Dimension	PubliBike (dock-based)	Pick-e-Bike (free-floating)
Power user rate	**23.5%**	**9.6%**
Dominant driver	**Weather resilience (RRR ≈ 2.78)**	**Multimodal integration (RRR ≈ 1.90)**
Secondary driver	**Functional superiority over PT**	**Functional superiority over PT**
Trip substitution	Primarily PT (76%)	PT + multimodal mix
Temporal pattern	AM/PM commuter peaks	Flatter; more weekend use (25% vs. 21%)

System types shape/attract different driver profiles.

Two archetypes, two system types

Routine Optimizer

Dock-based (PubliBike)

- Commutes to work on the same route
- High frequency, predictable timing
- Values station reliability
- EBSS as primary urban transport
- 76% substitute public transit trips

Flexibility Maximizer

Free-floating (Pick-e-Bike)

- Uses EBSS opportunistically
- Variable trips, many origins
- Values drop-anywhere freedom
- Combines EBSS with transit and car
- Key use: last-mile and mobility backup

System design shapes user type — or self-selects for it.

EBSS complements public transit — it does not replace the car

PT Substitute
76% of EBSS trips substitute public transit rides

Last-Mile Connector
Bridges gap between transit stations and destinations

Mobility Safety Net
Flexible backup when PT is delayed or unavailable

E-Bike Incubator
Trial experience accelerates private e-bike adoption

The key asymmetry:

Out of users with car access, only 34% substitute car trips

PT substitution by segment:

Power users: ∼76% of trips replace PT
Regular users: ∼70% replace PT
Occasional users: higher car substitution

PubliBike: stronger PT substitution

Pick-e-Bike: mixed PT + car + walk

For sustainability policy:

Carbon gains come from complementing and easing pressure on public transit — not from car displacement.

Three Evidence-Based Policy Recommendations

1 Design for Pragmatic Power Users

Don't: Focus marketing on environmental benefits alone
Do: Emphasize speed, reliability, PT connectivity
Action: Fare integration, peak-hour availability guarantees, station co-location/fleet adjustments

2 Operational Model Shapes User Cultivation

Free-Floating → Flexible "multimodalists" (PT access gaps, last-mile)
Dock-Based → Routine commuters (2.4× more power users)
Want car substitution? → difficult; expand service areas/station network
Want PT integration? → EBSS + PT fare bundles

3 Manage PT Substitution Strategically

PT substitution ≠ bad if: peak-hour capacity relief, users remain multimodal, improves first/last-mile access

Action: Co-locate stations at congested PT stops, time-of-use pricing, frame as "overflow valve"

Limitations and future research

Scope limitations:

- **Single region** — Basel's cycling culture may limit transferability
- **Self-selection bias** — active users overrepresented
- **MNL performance okay but not great** — shared mobility behavior complex to predict
- **No causal analysis** — observational design

Future Research:

- Longitudinal tracking of usage intensity transitions
- Experimental research with PT + EBSS fare bundles/seamless fare integration (WiP)

Thank you for your attention!

any questions?

Michael Stiebe (HSLU Luzern) · Benjamin Weggelaar (University of Basel)

Factor Items

F1 — Weather-Independent Usage Propensity

"How likely would you use e-bike sharing under the following weather conditions?"
(1 = Very unlikely . . . 5 = Very likely)

$\alpha = 0.813$

- High temperature (>30 °C) (Mob_Weather_1)
- Low temperature (<5 °C) (Mob_Weather_2)
- Light rain (Mob_Weather_3)
- Heavy rain (Mob_Weather_4)
- Wind (medium to strong) (Mob_Weather_5)
- No daylight / darkness (Mob_Weather_6)

F2 — Public Sector Intervention Attitudes

"Please rate the following statements regarding public support measures for e-bike sharing."
(1 = Strongly disagree . . . 5 = Strongly agree)

$\alpha = 0.532$

- Support public monetary investments in more bike lanes and better bike infrastructure (Policy_1)
- Support public monetary subsidies for e-bike sharing (equal to public transport) (Policy_2)
- Support non-monetary support (e.g. dedicated parking zones) (Policy_3)
- Support integration of e-bike sharing with existing public transport apps (e.g. SBB) (Policy_4)
- Support fully private financing of e-bike sharing (Policy_5, reverse-coded)

F3 — Hedonic Motivation & Social Support

"Please rate the extent to which you agree with the following statements."
(1 = Strongly disagree . . . 5 = Strongly agree)

$\alpha = 0.665$

- E-bike sharing is enjoyable (Environment_TPB_4)
- E-bike sharing is comfortable (Environment_TPB_3)
- My friends and family support the concept of e-bike sharing (Environment_TPB_5)
- I have enough experience to safely use an e-bike (Environment_TPB_6)

F4 — Perceived Functional Superiority over PT

"I use e-bike sharing because . . . "
(1 = Strongly disagree . . . 5 = Strongly agree)

$\alpha = 0.728$

- . . . it is more flexible than public transport (e.g. poor connections) (Mob_EBSS_Reasons_1)
- . . . it is more convenient than public transport (Mob_EBSS_Reasons_2)
- . . . it is faster than local public transport (Mob_EBSS_Reasons_3)
- . . . it is more fun than public transport (Mob_EBSS_Reasons_4)
- . . . it is healthier than public transport (Mob_EBSS_Reasons_5)

F5 — Multimodal Complementarity Behavior

"Please complete the following statements. I use e-bike sharing . . . "
(1 = Never . . . 5 = Very often)

$\alpha = 0.598$

- . . . for complete trips from A to B (Mob_Statements_1)
- . . . in combination with public transport (Mob_Statements_2)
- . . . in combination with walking (>5 min) (Mob_Statements_4)
- . . . as a fallback / backup option (Mob_Statements_5)

F6 — Service Reliability & Affordability Concerns

Included exploratorily — low internal consistency

$\alpha = 0.401$

- E-bike sharing is too expensive for regular use (Barr_Statements_2)
- Availability varies too much to depend on it (Barr_Statements_1)
- EBSS is only a fallback option (PT unavailable / bike not accessible) (Barr_Statements_4)
- How important are discounts or vouchers for your decision to use e-bike sharing? (Discount_Importance)

F7 — Environmental Self-Identity

"Please rate the extent to which you agree with the following statements."
(1 = Strongly disagree . . . 5 = Strongly agree)

$\alpha = 0.633$

- I think I have responsibility for reducing my environmental impact (Environment_TPB_1)
- Using e-bike sharing is good for alleviating environmental pollution (Environment_TPB_2)

EBSS as an e-bike incubator: trial experience can accelerate private adoption



Policy tension: effective incubation reduces the future EBSS user pool

- Private e-bike sales accelerated in Basel region alongside EBSS growth
- EBSS lowers the barrier to first e-bike experience
- Intensive users are most likely to transition to private ownership

Policy recommendation:

Position EBSS as an *on-ramp* to sustainable mobility, not just a destination. Subsidies for EBSS may indirectly stimulate the private e-bike market.

Design implication:

Track private e-bike purchase intent among members. High intent signals both a policy success and a risk of churn.

The most effective EBSS may shrink its own market — and that is a good outcome.